

A Counterexample to Conjecture 00000003965: Boxicity of $K_{2,2,2,2}$ and of Its Complement

Feiyu*

Claude (Opus 5.5)[†]

2026-10-03

Abstract

Conjecture 00000003965 asserts that every graph G satisfies $|\text{box}(G) - \text{box}(\overline{G})| \leq 2$. For the complete multipartite graph $G = K_{2,2,2,2}$ we have $\text{box}(G) = 4$, while its complement is a perfect matching with $\text{box}(\overline{G}) = 1$. So the difference is 3. The refutation is checked in Lean 4 against Mathlib.

1 The conjecture as stated

The original text of conjecture 00000003965 reads:

Definition: The boxicity of a graph is the minimal dimension in which the graph is represented as an intersection of interval boxes. Conjecture: Every graph G satisfies $|\text{box}(G) - \text{box}(\text{complement of } G)| \leq 2$ with respect to its complement, and the difference 2 is attained by an explicit construction.

The Chinese version in `conjectures/00000003965.md` states the same claim.

2 Reading of the statement

Definition 1. A d -dimensional box representation of a graph G assigns to every vertex v an axis-parallel box $\prod_{i=1}^d [l_i(v), r_i(v)] \subseteq \mathbb{R}^d$, with $l_i(v) \leq r_i(v)$, such that two distinct vertices are adjacent if and only if their boxes intersect. Two boxes intersect iff their intervals overlap in every coordinate, where $[a, b]$ and $[c, e]$ overlap iff $a \leq e$ and $c \leq b$. The *boxicity* $\text{box}(G)$ is the least d for which such a representation exists.

In dimension 0 every box is the single point $\mathbb{R}^0$, so exactly the complete graphs have boxicity 0; this is the usual convention. The first claim of the conjecture is $|\text{box}(G) - \text{box}(\overline{G})| \leq 2$ for every graph G , where $\overline{G}$ is the complement. We refute it with a finite graph, so restricting the claim to finite graphs makes no difference. We do not need the second claim.

3 Result

Let $G = K_{2,2,2,2}$ be the complete multipartite graph with four parts of size two. Its vertices are the pairs (i, s) with $i \in \{1, 2, 3, 4\}$ and $s \in \{0, 1\}$, and (i, s) is adjacent to (k, t) iff $i \neq k$. The complement $\overline{G}$ is the perfect matching $4K_2$ with edges $\{(i, 0), (i, 1)\}$.

*Independent researcher.

[†]Anthropic.

Theorem 2. $\text{box}(G) = 4$ and $\text{box}(\overline{G}) \leq 1$. Hence $\text{box}(G) - \text{box}(\overline{G}) \geq 3$, and Conjecture 00000003965 is false.

This is the case $k = 4$ of Roberts' theorem that the complete multipartite graph with k parts of size two has boxicity k [1]. We give the short proof.

4 Proof

Lemma 3. Let A, B, C, E be closed intervals of $\mathbb{R}$ such that $A \cap B = \emptyset$ and each of C and E meets both A and B . Then $C \cap E \neq \emptyset$.

Proof. Say $A = [a_1, a_2]$ lies to the left of $B = [b_1, b_2]$, so $a_2 < b_1$. Since $C = [c_1, c_2]$ meets A and B , we have $c_1 \leq a_2$ and $c_2 \geq b_1 > a_2$, so $a_2 \in C$. Likewise $a_2 \in E$. $\square$

Proof of Theorem 2. Upper bound for $\overline{G}$. Give $(i, 0)$ and $(i, 1)$ the interval $[3i, 3i + 1]$. Intervals of the same i coincide, and intervals of different i are disjoint, so this is a 1-dimensional representation of $\overline{G}$.

Upper bound for G . In coordinate $j \in \{1, 2, 3, 4\}$ give $(j, 0)$ the interval $[0, 1]$, give $(j, 1)$ the interval $[2, 3]$, and give every vertex (i, s) with $i \neq j$ the interval $[0, 3]$. In coordinate j the only non-overlapping pair is $\{(j, 0), (j, 1)\}$. Hence two distinct vertices have intersecting boxes iff they do not form one of the pairs $\{(j, 0), (j, 1)\}$, that is, iff they are adjacent in G . So $\text{box}(G) \leq 4$.

Lower bound for G . Suppose G has a d -dimensional representation with $d \leq 3$. The vertices $(i, 0)$ and $(i, 1)$ are distinct and non-adjacent, so their boxes are disjoint and there is a coordinate $\text{sep}(i)$ in which their intervals are disjoint. Since sep maps four values into $d \leq 3$ coordinates, $\text{sep}(i) = \text{sep}(k) = j$ for some $i \neq k$. In coordinate j the intervals A, B of $(i, 0), (i, 1)$ are disjoint. The intervals C, E of $(k, 0), (k, 1)$ each meet A and B , because (k, t) is adjacent to (i, s) . By Lemma 3, C and E meet, contradicting the choice of $j = \text{sep}(k)$. Hence $\text{box}(G) \geq 4$. $\square$

5 Formalization

The Lean 4 project in `lean/` (file `lean/Submission/Basic.lean`) formalizes the definitions above and proves Theorem 2 against Mathlib.

- `BoxRep G d` is a d -dimensional box representation: real endpoint maps l, r on vertices and coordinates $i \in \text{Fin } d$ with $l_i(v) \leq r_i(v)$, such that distinct u, v are adjacent iff $l_i(u) \leq r_i(v)$ and $l_i(v) \leq r_i(u)$ for all i . `boxicity G` is the infimum (`sInf`) of the dimensions admitting one.
- `BoxicityGapAtMostTwo` states $|\text{box}(G) - \text{box}(\overline{G})| \leq 2$ for every finite graph, with $\overline{G}$ Mathlib's complement of G .
- The graph G is Mathlib's `completeEquipartiteGraph 4 2` on `Fin 4 × Fin 2`.
- `repCompl` and `repK` are the two representations above; the adjacency conditions are checked by `decide` on integer endpoints.
- `interval_lemma` is Lemma 3, and `no_rep_of_lt_four` is the lower bound, using the pigeonhole principle `Fintype.exists_ne_map_eq_of_card_lt`.
- `boxicity_K` gives $\text{box}(G) = 4$, `boxicity_compl_le` gives $\text{box}(\overline{G}) \leq 1$, and `conjecture_00000003965_false` is $\neg \text{BoxicityGapAtMostTwo}$.

The toolchain is `leanprover/lean4:v4.33.1`, Mathlib is pinned to `v4.33.1`, and `lake build` succeeds. The project contains no `sorry`, no `admit` and no `native_decide`, and introduces no axioms: `#print axioms` reports exactly `propext`, `Classical.choice` and `Quot.sound` for `conjecture_00000003965_false` and `boxicity_K`.

6 Remarks

Remark 4. The same construction gives unbounded differences: for the complete multipartite graph with k parts of size two, $\text{box} = k$ while the complement (a perfect matching) has boxicity 1 [1]. So $|\text{box}(G) - \text{box}(\overline{G})|$ is not bounded by any constant.

References

- [1] F. S. Roberts, *On the boxicity and cubicity of a graph*, in: Recent Progress in Combinatorics (W. T. Tutte, ed.), Academic Press, 1969, 301–310.